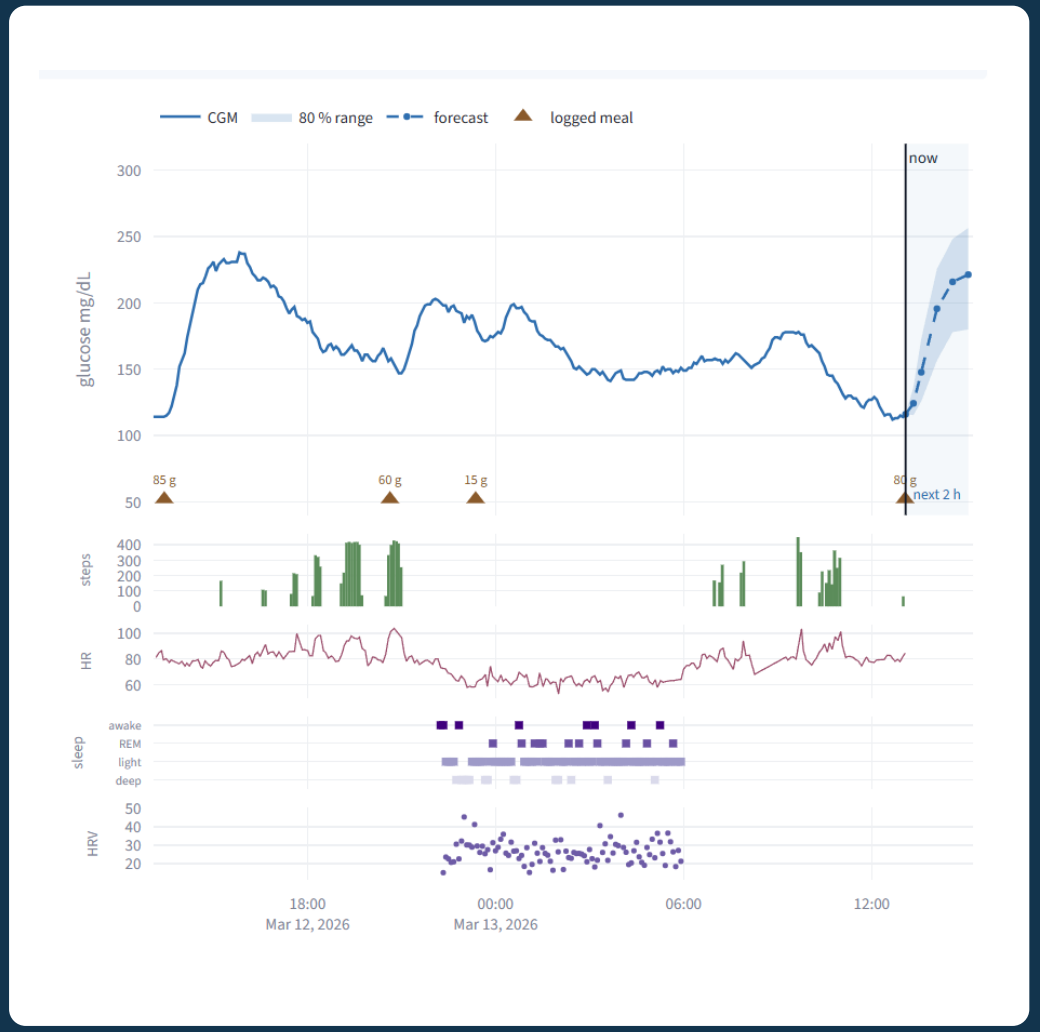
DIGITAL TWIN CHALLENGE 2026 · HAPPIEST HEALTH

A digital twin for Type 2 diabetes

Predicts a glucose spike about 2 hours ahead, explains why in clinical language, and lets a doctor test what-ifs on a personalised virtual patient.



Team AMRIT_VIT
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Synthetic patients only. Decision support prototype, not a medical device.

The problem: plenty of glucose data, a few minutes per patient

- India has an estimated 101 million adults with diabetes (ICMR-INDIAB, 2023). Type 2 diabetes often starts younger and at a lower BMI than in Western cohorts.
- HbA1c is a three-month average: it cannot show when, or why, glucose spikes after today's meals.
- A CGM gives 288 readings a day, and a wrist wearable adds heart rate, HRV, sleep and steps. Nobody can read all of it in a short consultation.
- Real patient data cannot be freely shared to build AI (DPDP Act 2023), so development needs realistic synthetic patients.

Healthcare use case
A diabetologist following CGM-wearing Type 2 patients, in clinic or remotely: who needs attention in the next 2 hours, why, and which change to discuss (a meal, a walk, or a medicine).

101 M

adults with diabetes in India (ICMR-INDIAB, Lancet Diabetes Endocrinol 2023)

4,032

CGM readings per patient in two weeks (one every 5 minutes)

0

real patient records used to build or train anything here

What we built: a twin that predicts, explains and simulates

Static record (EHR)

Age, sex, BMI, waist · HbA1c and its 12-month trend · fasting glucose · eGFR, ALT, fatty liver · lipids, BP · medicines with dose and adherence · allergies and intolerances · diet (vegetarian, non-vegetarian, Jain)

Dynamic streams

CGM every 5 minutes · heart rate and HRV · sleep stages · steps · the patient's own meal log (optional, often incomplete)



1

Predict

Calibrated probability that glucose goes above 180 mg/dL within 2 hours, plus a 15-120 minute forecast with an 80 % band. Model: LightGBM fusion of all streams.

2

Explain

SHAP contributions to the predicted peak, grouped into clinical concepts and written as sentences in mg/dL, compared with similar patients.

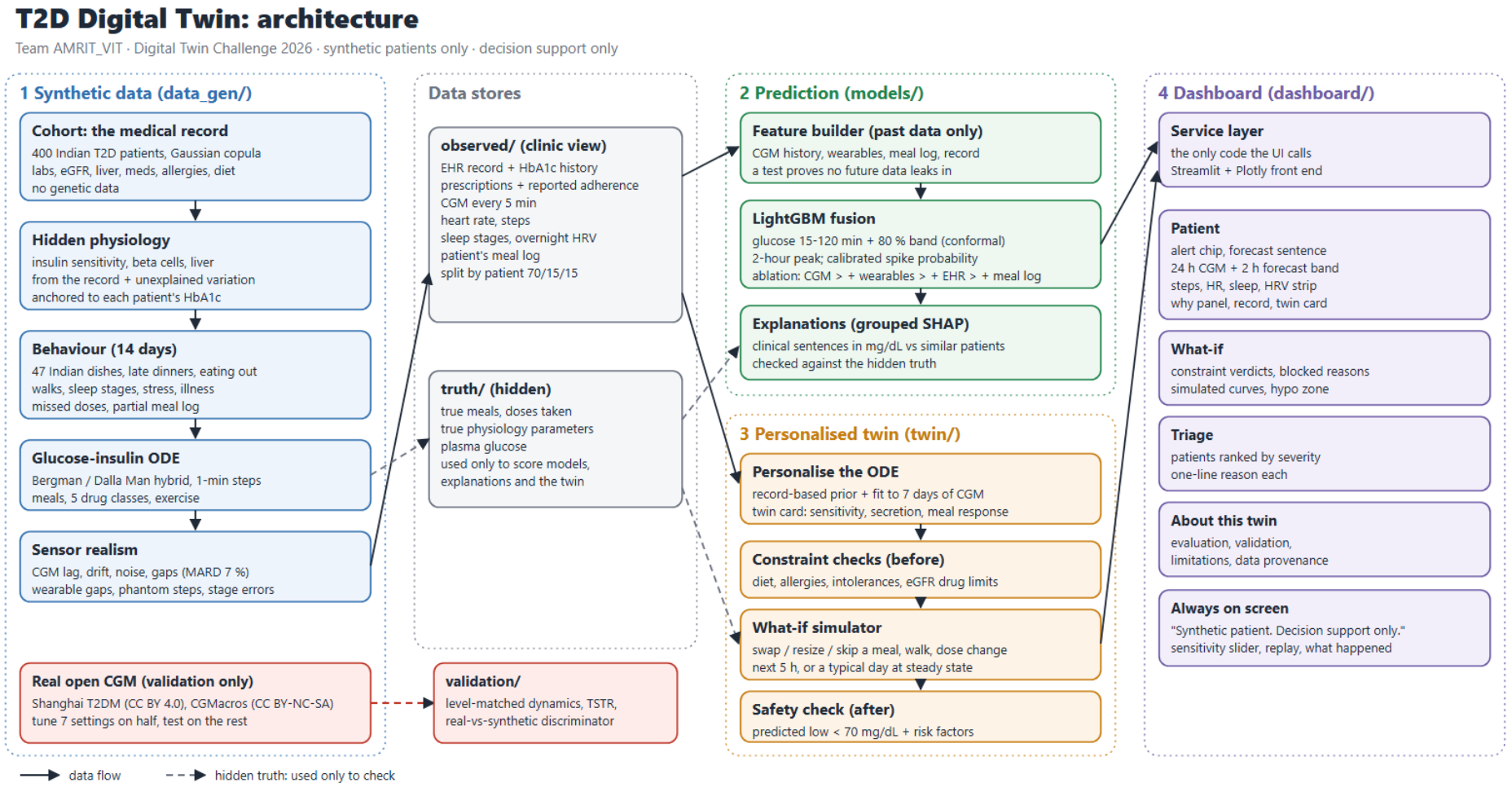
3

Simulate

A personalised glucose-insulin ODE twin answers "what if?" for meals, walks and medicines. Constraints are checked first; every result gets a hypoglycaemia check.

All three are shown to the doctor in one dashboard: Patient · What-if · Triage · About this twin.

Architecture



Held-out results: 92 % of spikes caught, median 80 min ahead, 0.69 false alerts/patient-day (threshold 0.5) · what-if dish swap error 13 mg/dL vs 31 for the cohort average
train-on-synthetic, test-on-real 60-min error 18.7 vs 19.9 mg/dL (no-change baseline)

Realistic synthetic patients, built in four layers

1

Cohort (the record)

Gaussian copula over age, BMI, waist, HbA1c, fasting glucose, eGFR, lipids and BP with Indian-clinic distributions; allergies, diet type, eGFR-aware prescribing. No genetic fields.

2

Hidden physiology

Insulin sensitivity, beta-cell function and liver glucose output derived from the record plus variation the record cannot explain; a Bergman / Dalla Man glucose-insulin ODE with drugs and exercise.

3

Behaviour

47 Indian plates (idli-sambar, rajma chawal, poha...), late dinners, weekend shifts, walks, sleep, stress and illness days, missed doses, and a partial, biased meal log.

4

Sensors

CGM lag of 5-15 min, noise, drift, warm-up, dropouts and compression lows; wearable gaps, phantom steps and noisy HRV.

400 × 14

patients × days, split 70/15/15 by patient

1.6 M

CGM readings, 2.8 % missing

7.3 %

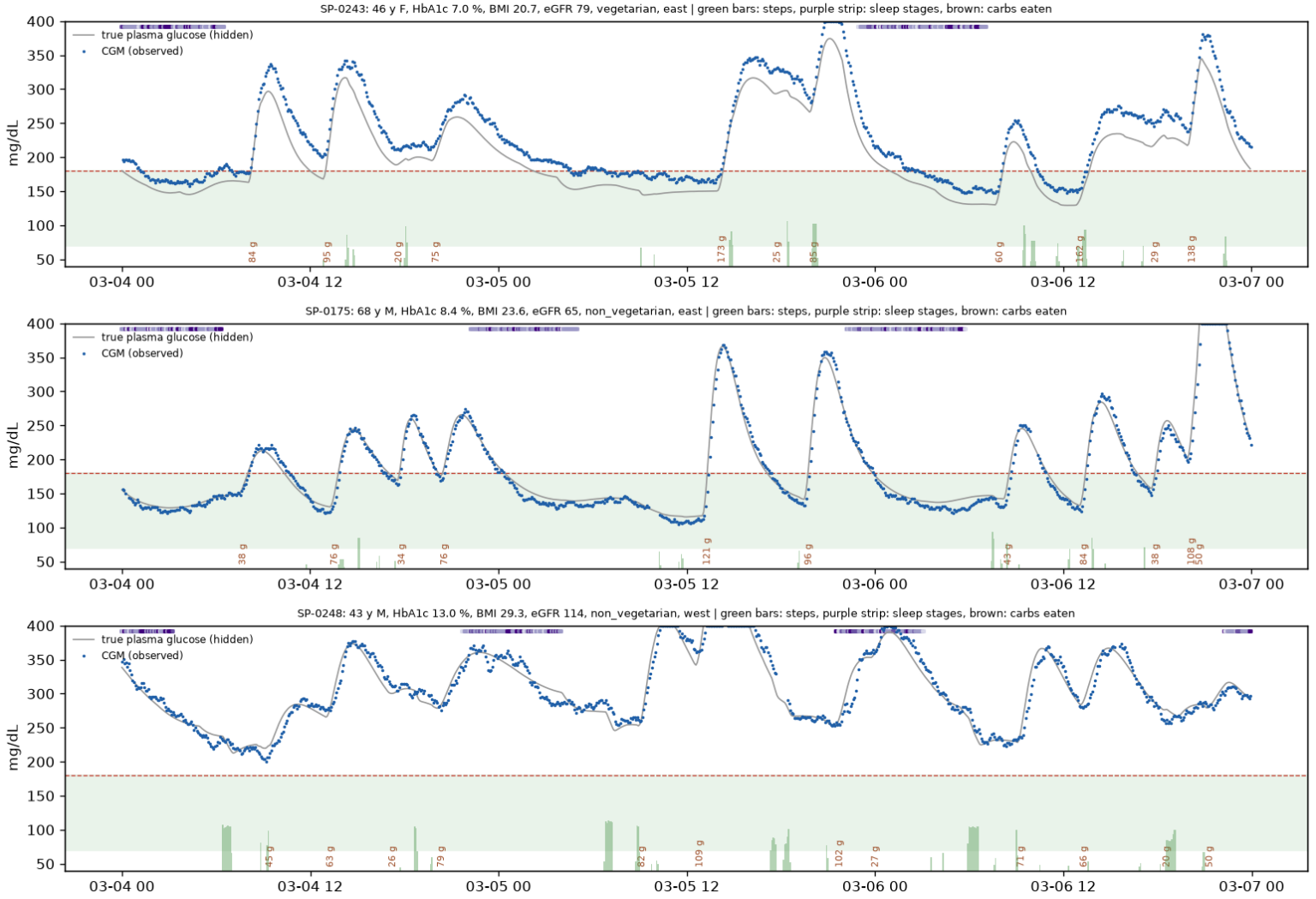
CGM error (MARD) vs the hidden true glucose

8.3 %

median HbA1c, 45 % time in range

Every patient regenerates identically from one seed. The hidden truth (true glucose, meals eaten, doses taken) is stored apart from what a clinic would see, and tests check nothing hidden leaks into the models.

What the generator produces: three patients, three days



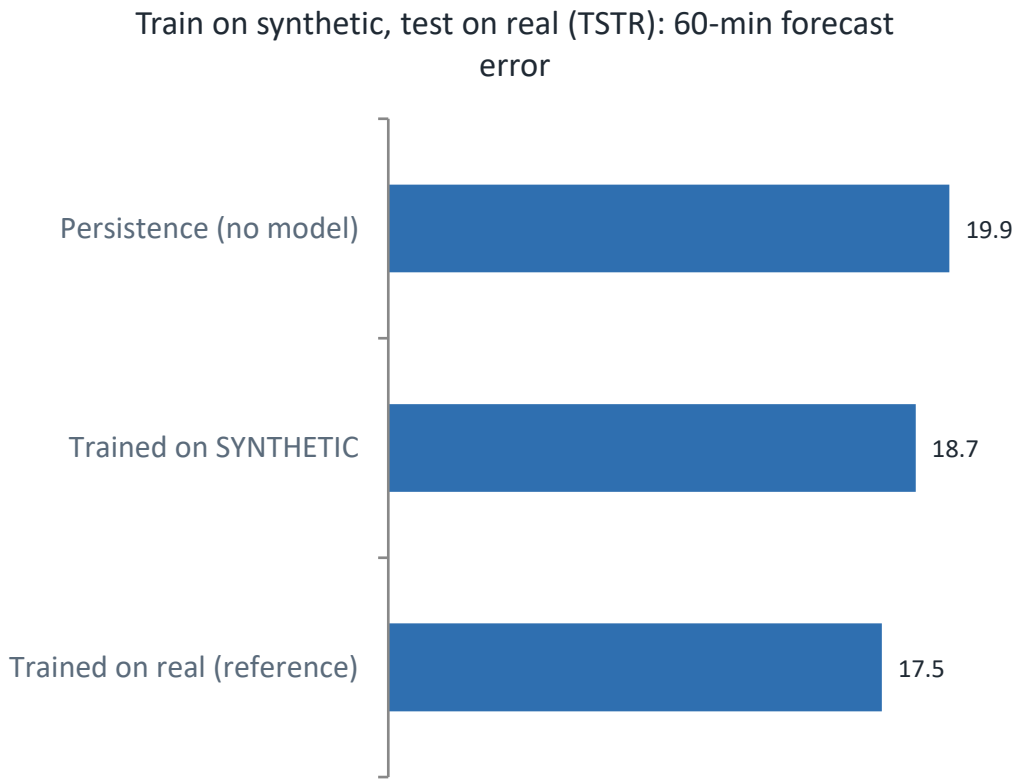
- **Grey:** true plasma glucose, hidden from every model.
- **Blue:** what the CGM reports, with lag, noise and gaps.
- **Brown:** carbohydrate eaten. Green: steps. Purple: sleep.
- The same meal gives very different curves at HbA1c 7.0, 8.4 and 13.0 %.
- Because we know the truth, we can test the models and the twin in ways real data never allows.

Checked against real Type 2 CGM data

metric (held-out Shanghai, 49 patients)	real	synthetic
coefficient of variation	0.27	0.26
night / mean glucose	0.82	0.82
time to post-meal peak	90 min	93 min
spikes above 180 per day	1.63	1.60
MAGE (outside the band)	75	92
fastest 15-min fall, p5 (outside)	−11	−14

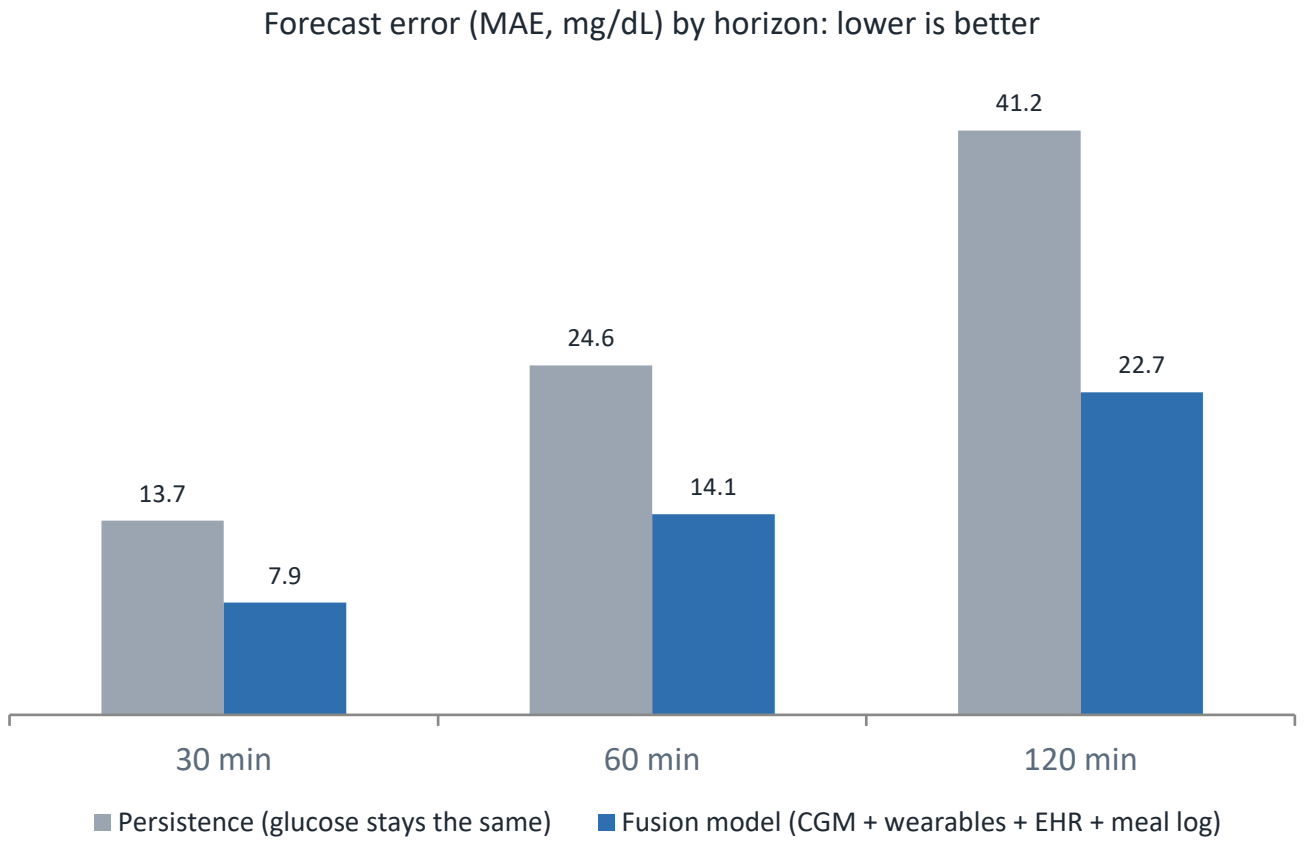
12 of 14 metrics fall within half an interquartile range of the real median. Real CGM runs lower than HbA1c implies (Shanghai: mean 142 vs HbA1c-estimated 194 mg/dL), so each real patient is compared with the synthetic patient of closest mean glucose.

Where it still differs: a classifier can tell real from synthetic 12-hour windows (AUC 0.95), and real glucose falls faster after peaks (−18 to −20 vs −15 mg/dL per 15 min).



A CGM-only model trained only on synthetic patients beats persistence on real Shanghai patients and comes close to a model trained on real data. Real data calibrates 7 settings and validates; it never trains the product models.

Forecast and spike detection on held-out patients



60 test patients never seen in training (230,609 prediction times). Spike = CGM above 180 mg/dL for at least 15 minutes within the next 2 hours.

14.1
mg/dL error at 60 min (persistence 24.6)

99.6 %
of 60-min forecasts in Clarke zones A+B

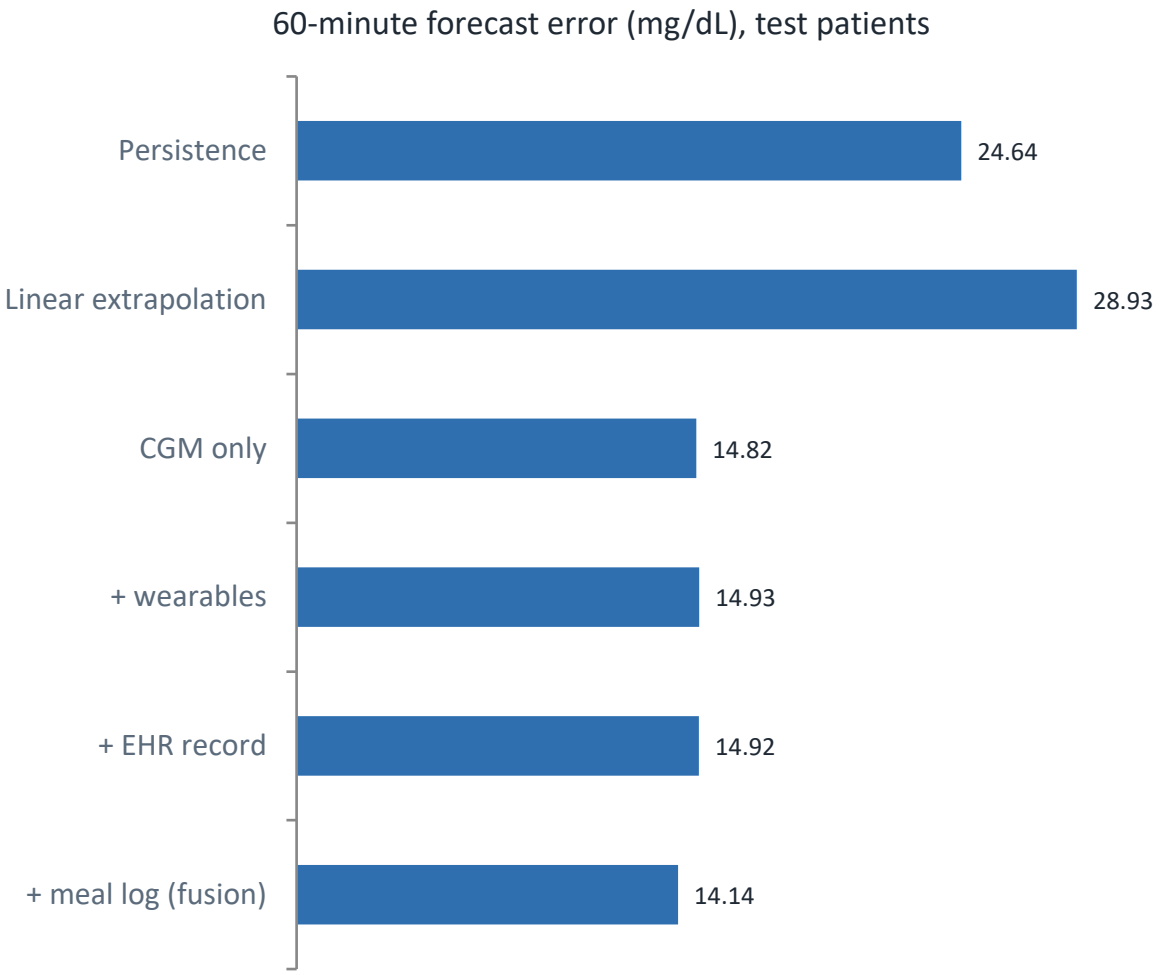
0.848
spike PR-AUC (AUROC 0.915; spike rate 35 %)

0.115
Brier score after isotonic calibration

79 %
of outcomes inside the 80 % band

80 min
median warning before a spike (threshold 0.5)

Does fusing the streams help? An honest ablation



1 Meal log

Clear gain: 60-min error −0.78 mg/dL [−0.90, −0.64], PR-AUC +0.010.

2 Wearables

Help where physiology says they should: right after a walk, −0.90 mg/dL [−1.64, −0.24]. Slightly worse overall (+0.11).

3 EHR record

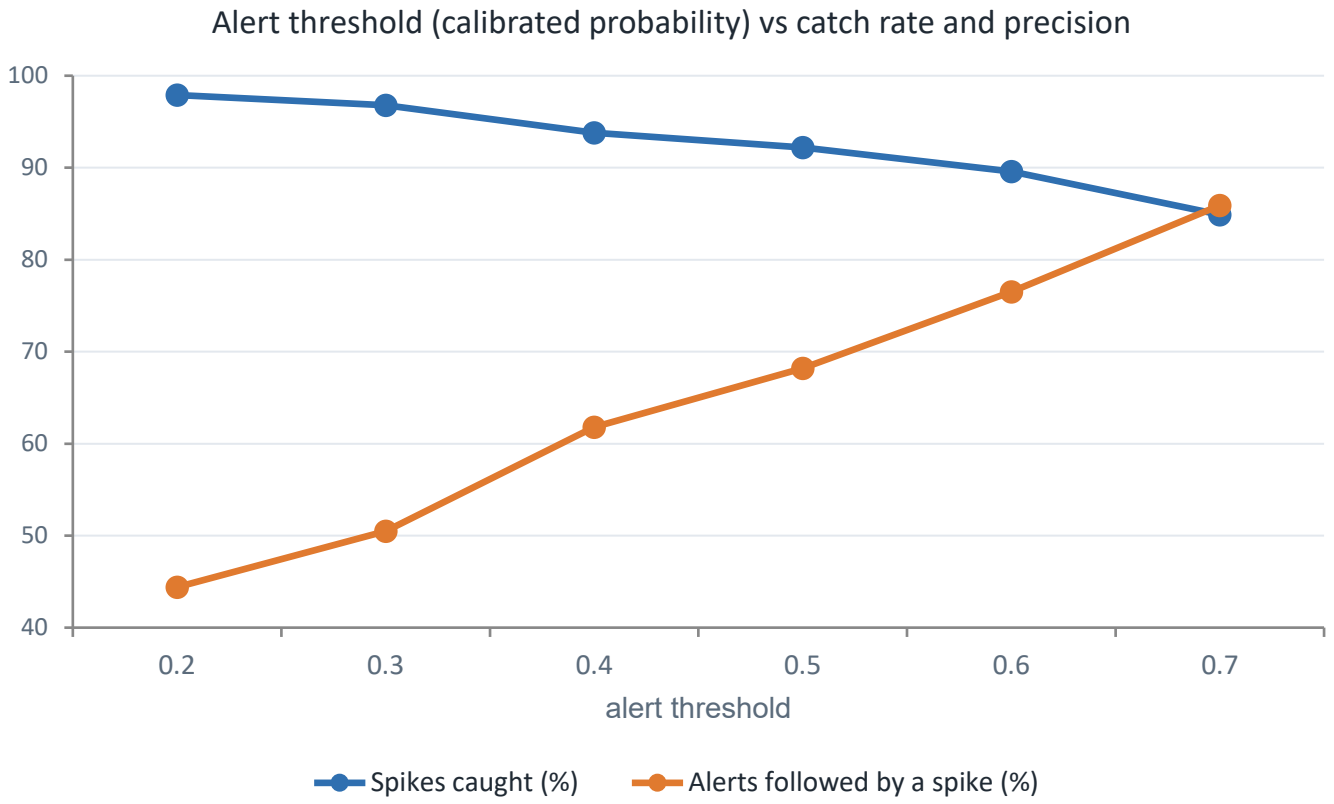
Adds PR-AUC +0.010 in a patient's first 24 h, when CGM history is short. No overall effect.

4 No meal log

Robust to messy inputs: CGM + wearables alone still reach 14.9 mg/dL (persistence 24.6).

Paired differences with patient-level bootstrap 95 % intervals. With hours of CGM history, the trace already carries most of today's physiology.

Alerts a clinic can live with



threshold	0.2	0.3	0.4	0.5	0.6	0.7
false alerts / patient-day	1.32	1.15	0.85	0.69	0.49	0.27

Dashboard default: threshold 0.5

92 %
of spikes caught

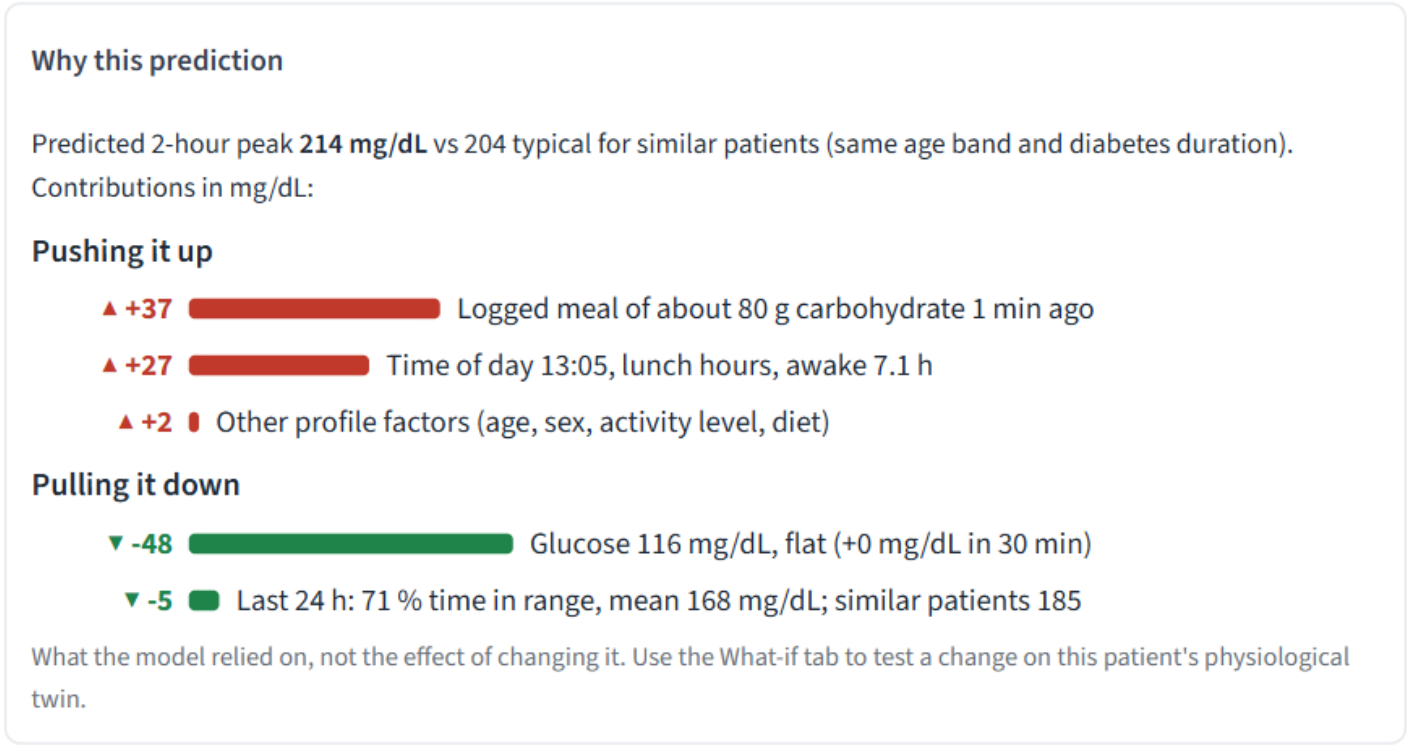
80 min
median warning time

0.69
false alert episodes per patient-day

68 %
of alerts followed by a spike

The doctor's sensitivity slider moves along this curve: at 0.36, 96 % of spikes are caught at 1.0 false alert a day; at 0.7, 85 % at 0.27 a day. Fewer false alarms means less alert fatigue.

Why this alert: explanations a doctor can read



Scene 1 (SP-0218, 13:05): a meal logged a minute ago pushes the predicted peak up; her current glucose of 116 pulls it down.

99.8 %

after a logged meal of 60 g or more, "logged meals" is a top-3 driver

19.5 vs 5.6

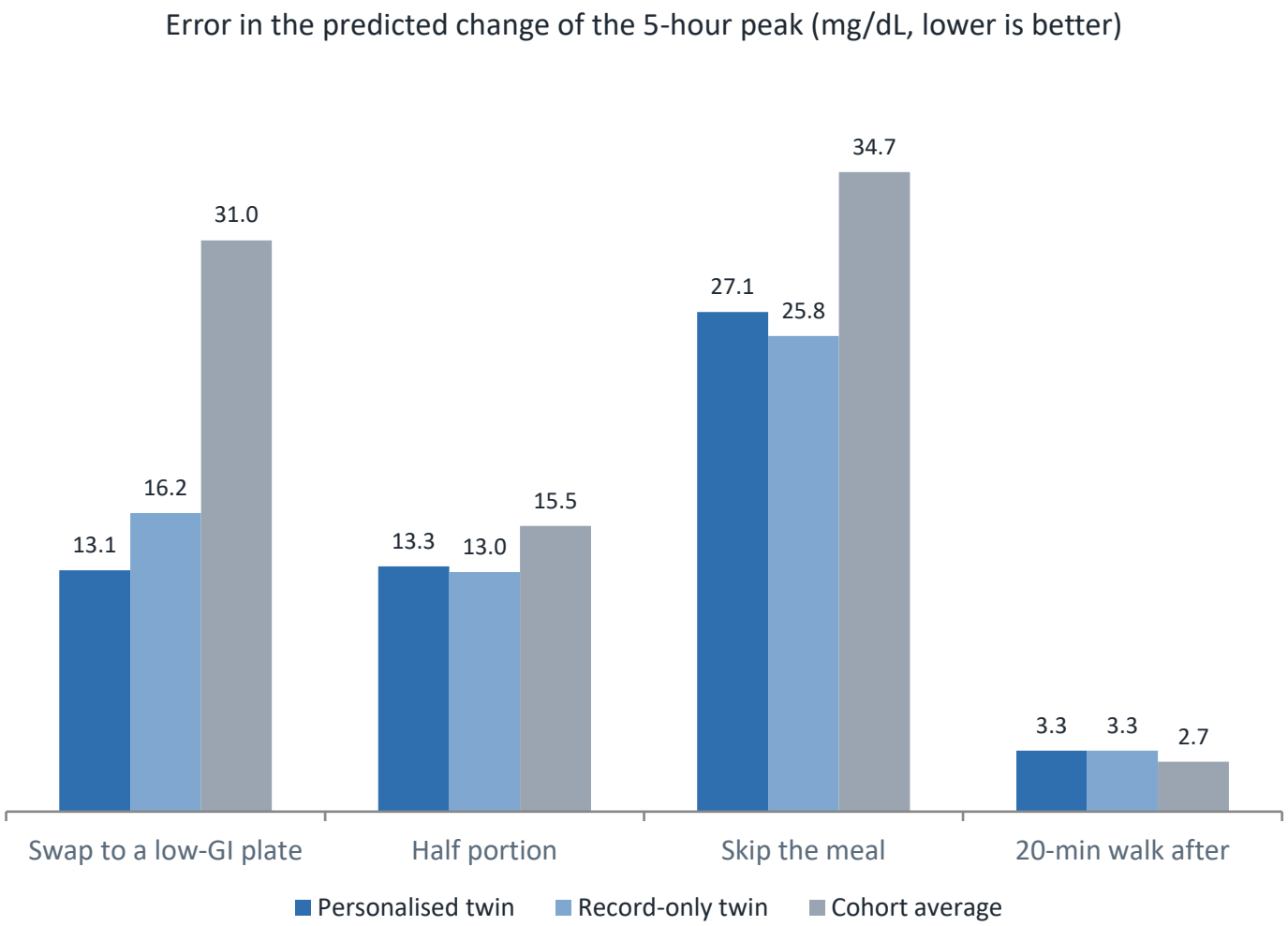
mg/dL moved by removing the top-3 concepts vs 3 random ones (deletion test)

0.75 vs 0.04

explanation similarity for similar moments vs random pairs

SHAP shows what the model relied on, never the effect of a change. What-if answers come only from the ODE twin.

What-if answers, checked against the truth



- Only synthetic data allows this test: each of 50 test patients is replayed with the change applied to their true meals, doses and activity.
- True median effects: swap −34, half portion −45, skip −84, walk −5 mg/dL.
- The predicted change has the right direction for 94 % of patients (84 % for the swap). Personalisation helps most for dish swaps (Spearman 0.91).
- Both twins beat the cohort average for every meal change. Large cuts are under-estimated, and a walk's small effect is not predictable.

Constraints first, a safety check always

1

Check

Diet type, allergies, intolerances and eGFR drug limits are checked before anything is simulated. Blocked options show why.

2

Simulate

Allowed options run on the patient's personalised twin for the next 5 hours, or as a typical day for a medicine change.

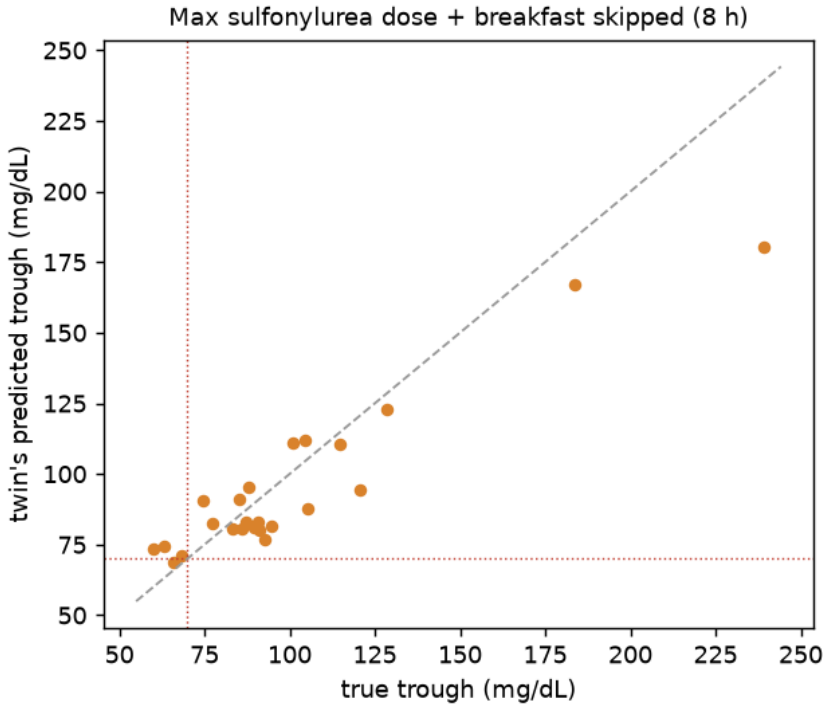
3

Flag

A predicted trough below 70 mg/dL is flagged as hypoglycaemia (below 80: caution), with the patient's risk factors.

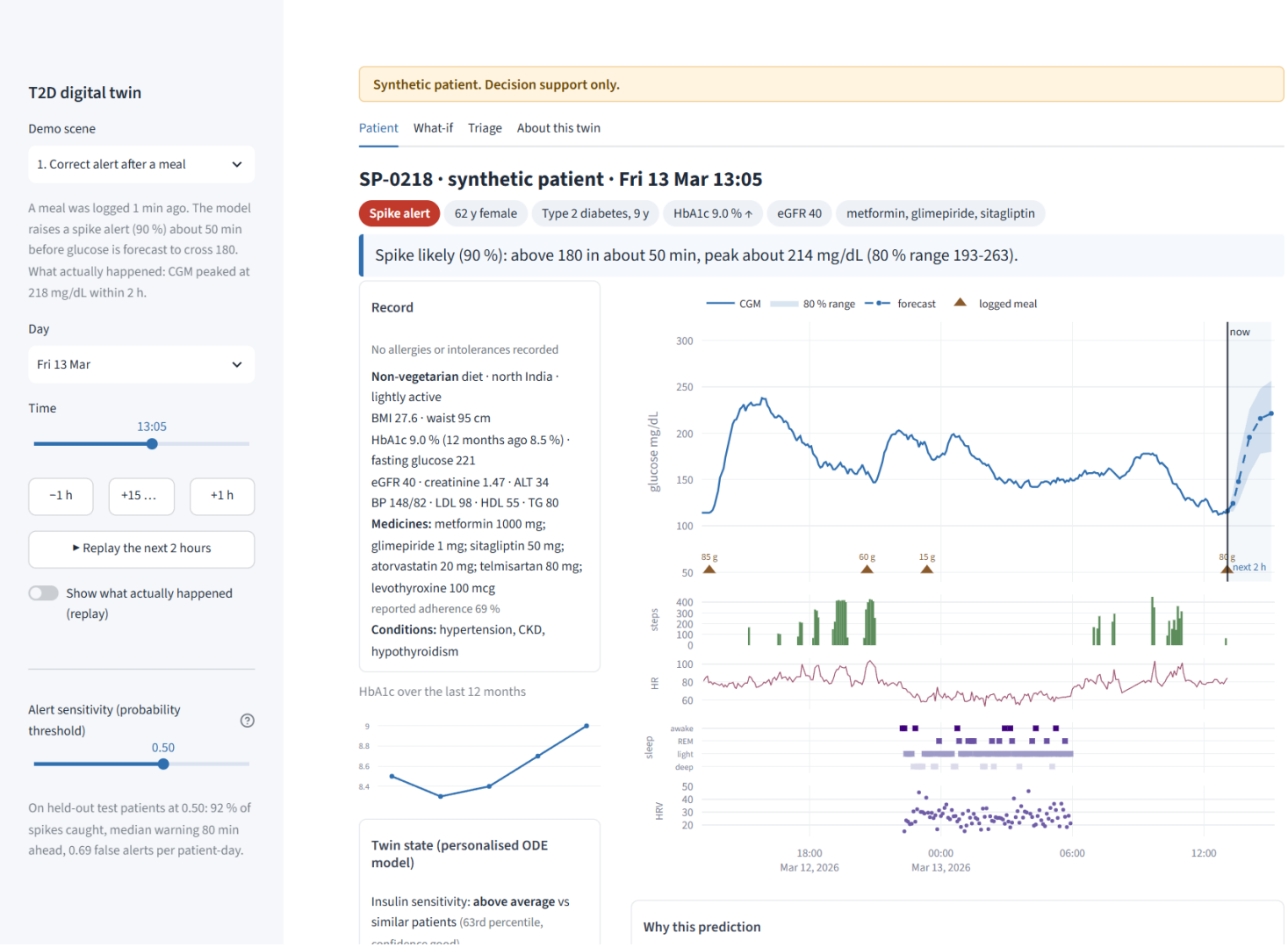
⛔ **Metformin 1000 → 2000 mg/day: BLOCKED. eGFR 40 (30-44): metformin limited to 1000 mg/day.**

Every result is labelled "Simulation for discussion with the patient, not a prescription". The rules are listed for clinical review (docs/clinical_rules_to_review.md).



Stress test on 24 sulfonylurea users (maximum dose, breakfast skipped): all 4 true lows were flagged at least "caution", 1 as a predicted low, with no false alarms.

The doctor's screen: one patient at a glance



1. **Alert chip and one sentence**
   "Spike likely (90 %): above 180 in about 50 min, peak about 214 mg/dL (80 % range 193-263)."

2. **24 h CGM + 2 h forecast**
   Target range, spike line, logged meals and the 80 % band.

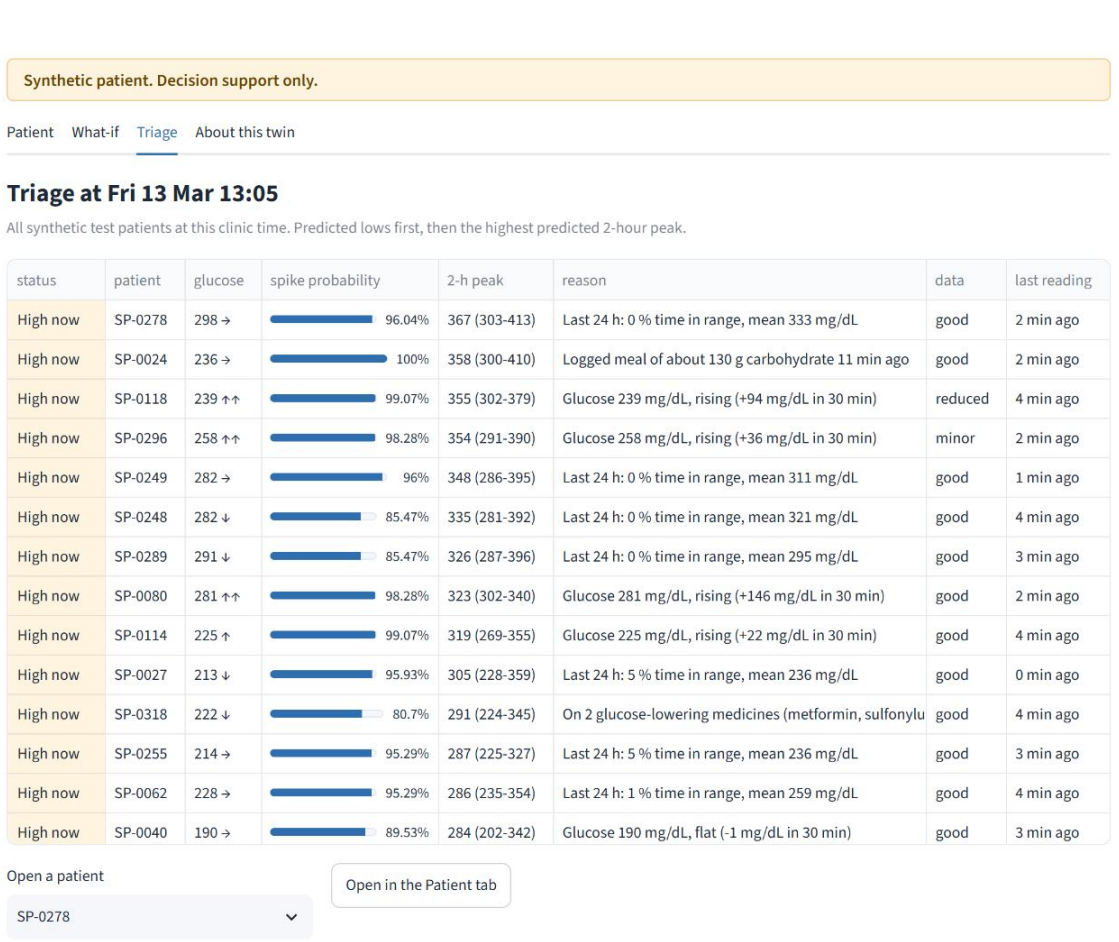
3. **Wearables on the same time axis**
   Steps, heart rate, sleep stages and HRV.

4. **Record card**
   HbA1c trend, eGFR, medicines, allergy badges; the twin's state below.

5. **Sensitivity slider and replay**
   Replay steps the twin forward; "show what actually happened" for the demo.

What-if on the twin, and triage across patients





status	patient	glucose	spike probability	2-h peak	reason	data	last reading
High now	SP-0278	298 →	96.04%	367 (303-413)	Last 24 h: 0 % time in range, mean 333 mg/dL	good	2 min ago
High now	SP-0024	236 →	100%	358 (300-410)	Logged meal of about 130 g carbohydrate 11 min ago	good	2 min ago
High now	SP-0118	239 ↑↑	99.07%	355 (302-379)	Glucose 239 mg/dL, rising (+94 mg/dL in 30 min)	reduced	4 min ago
High now	SP-0296	258 ↑↑	98.28%	354 (291-390)	Glucose 258 mg/dL, rising (+36 mg/dL in 30 min)	minor	2 min ago
High now	SP-0249	282 →	96%	348 (286-395)	Last 24 h: 0 % time in range, mean 311 mg/dL	good	1 min ago
High now	SP-0248	282 ↓	85.47%	335 (281-392)	Last 24 h: 0 % time in range, mean 321 mg/dL	good	4 min ago
High now	SP-0289	291 ↓	85.47%	326 (287-396)	Last 24 h: 0 % time in range, mean 295 mg/dL	good	3 min ago
High now	SP-0080	281 ↑↑	98.28%	323 (302-340)	Glucose 281 mg/dL, rising (+146 mg/dL in 30 min)	good	2 min ago
High now	SP-0114	225 ↑	99.07%	319 (269-355)	Glucose 225 mg/dL, rising (+22 mg/dL in 30 min)	good	4 min ago
High now	SP-0027	213 ↓	95.93%	305 (228-359)	Last 24 h: 5 % time in range, mean 236 mg/dL	good	0 min ago
High now	SP-0318	222 ↓	80.7%	291 (224-345)	On 2 glucose-lowering medicines (metformin, sulfonylu	good	4 min ago
High now	SP-0255	214 →	95.29%	287 (225-327)	Last 24 h: 5 % time in range, mean 236 mg/dL	good	3 min ago
High now	SP-0062	228 →	95.29%	286 (235-354)	Last 24 h: 1 % time in range, mean 259 mg/dL	good	4 min ago
High now	SP-0040	190 →	89.53%	284 (202-342)	Glucose 190 mg/dL, flat (-1 mg/dL in 30 min)	good	3 min ago

What-if: patient factors as chips, the current plan vs the change, and the hypoglycaemia check.

Triage: every patient at one clinic time, predicted lows first, then the highest 2-hour peak, with a one-line reason and data quality.

Responsible data use and honest limitations

Responsible data (DPDP-aware)

- No personal data anywhere: every patient is simulated.
- Real open datasets are downloaded by script, never committed, and used only to calibrate and validate (licences respected; CGMacros is non-commercial).
- Hidden simulator truth is stored apart; tests check nothing hidden leaks into model inputs.
- Every literature value carries a source and a "verified" flag (43 still to verify).
- "Synthetic patient. Decision support only." on every screen. No genetic markers, by design.

Limitations

- Models trained and scored on simulated patients can partly rediscover the simulator's assumptions; scores are an upper bound for real patients.
- The real validation sets are small and not Indian, and a classifier still separates real from synthetic windows (AUC 0.95).
- A week of CGM does not pin down the twin's parameters exactly, so the twin card shows its uncertainty.
- Indian population values, meal macros and drug rules await expert review before any clinical use.

Details: docs/data_card.md (circularity), docs/validation_report.md, docs/parameters_to_verify.md, docs/clinical_rules_to_review.md.

Impact and next steps

What it enables

- Triage of CGM-monitored patients by 2-hour risk, with a reason for each.
- Alerts about 80 minutes ahead, explained in clinical language.
- Meal, walk and medicine changes discussed on the patient's own twin, with Indian foods and safety checks.
- Privacy-safe development: realistic synthetic patients instead of real records.

Next steps

- Validate on an Indian CGM cohort; clinician review of rules and sentences.
- Prandial and premixed insulin, GLP-1 receptor agonists.
- Patient-side nudges (WhatsApp / SMS) and regional-language summaries.
- A silent-mode prospective study before any clinical use.

github.com/shashi-bhushan-27/digitalTwinChallenge

Team AMRIT_VIT · Shashi Bhushan · VIT Vellore · Thank you